

Research Paper 14 Summary

XGBoost: A Scalable Tree Boosting System

Problem Statement: AI-Based Detection and Classification of Industrial Fires (PS26162)

Original Paper: Chen & Guestrin (2016), *ACM SIGKDD*

1 Basic Information

- **Paper Title:** XGBoost: A Scalable Tree Boosting System
- **Authors:** Tianqi Chen and Carlos Guestrin (2016)
- **Conference:** *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*
- **Core Contribution:** Created ****XGBoost (Extreme Gradient Boosting)****, the world's leading machine learning algorithm for tabular and structured datasets.

2 What is this Paper About? (In Simple Words)

This paper introduces XGBoost, a high-performance machine learning system built on gradient boosted decision trees.

- **How Gradient Boosting Works:** Unlike Random Forest which trains multiple independent trees at once, XGBoost builds a sequence of trees step-by-step. Each new tree specifically focuses on correcting the errors made by the previous trees.
- **Built-in Regularization:** It adds mathematical penalties (L_1 and L_2 regularization) to prevent trees from becoming overly complex, stopping the model from memorizing noise.
- **Handling Missing Data Automatically:** In satellite imagery, clouds or sensor glitches often cause missing data values. XGBoost automatically learns the best default decision path when a data field is blank.

3 Key Concept: Handling Severe Class Imbalance

In real-world satellite fire monitoring across India, the data is heavily imbalanced:

For every 1 true industrial fire detection, there are over 100 agricultural stubble burns or natural forest fire alerts.

Standard AI models fail on imbalanced data by simply guessing the majority class (predicting everything is a farm fire). XGBoost includes built-in weighted gradient loss functions (`scale_pos_weight`) that force the algorithm to give high priority to rare industrial fire events.

4 How This Helps Our Project (PS26162)

4.1 1. The Primary AI Classification Engine for Our Solution

XGBoost serves as our core machine learning model to classify each NASA FIRMS detection into one of our target classes:

$$\hat{y} \in \{\text{Industrial Flare / Factory, Abnormal Industrial Fire, Wildfire, Agricultural Burn, Non-Fire Anomaly}\} \quad (1)$$

4.2 2. Solving the 1:100 Imbalance in Indian Satellite Feeds

By tuning the class-weight parameters in XGBoost, our model maintains high sensitivity (recall) for industrial fires without getting drowned out by the millions of seasonal stubble burning alerts across Northern India.

4.3 3. Ultra-Fast Inference for Real-Time Map Overlays

NASA FIRMS delivers thousands of new coordinates across India every few hours. XGBoost can process tens of thousands of detection rows in under 50 milliseconds, allowing our GIS map overlay to update in real time without lag.

4.4 4. Seamless Compatibility with SHAP for Explainability

XGBoost features native, highly optimized C++ bindings with TreeSHAP (from Paper 13), enabling our system to generate instant explainability scores for our user dashboard.